

MOMCHIL E. MOLNAR

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EMPLOYMENT

Southwest Research Institute, Boulder, Colorado

Research Scientist, October 2024 – present

I study the solar transition region with the SoLO/SPICE instrument, develop novel coronal magnetic field diagnostics, and advance physics-informed machine learning methods for spectropolarimetry.

High Altitude Observatory, NCAR, Boulder

Advanced Study Program Postdoctoral Fellow, October 2022 – October 2024

I worked on magnetic field diagnostics of the solar atmosphere based on the Hanle effect and novel spectropolarimetric inversion tools based on neural fields. PI of the CHEESE experiment to search for coronal He I 1083 nm emission during the April 8 2024 eclipse.

LASP, University of Colorado / National Solar Observatory, Boulder

Graduate Research Assistant, Fall 2016 – August 2022

I worked with solar observations from the DST, ALMA, IRIS, and Hi-C 2.1 combined with radiative transfer calculations to test wave heating theories of the solar chromosphere.

Kavli Institute for Astrophysics and Space Research, MIT, Cambridge

Undergraduate Research Assistant (UROP Program), Spring 2013 – Spring 2016

Topics I worked on included stellar pulsations, evolution of cataclysmic variable stars, and data mining.

EDUCATION

University of Colorado, Boulder

November 2018 – August 2022

PhD, Astrophysical and Planetary Sciences

Department of Astrophysical and Planetary Sciences

University of Colorado, Boulder

August 2016 – October 2018

MSc, high-pass distinction

Department of Astrophysical and Planetary Sciences

Massachusetts Institute of Technology

September 2012 – June 2016

BSc, Physics

Undergraduate Thesis: Automatic detection of periodic sources in the K2 data sets

PEER-REVIEWED PUBLICATIONS

1. Jarolim, R., Molnar, M., ..., “Spherical PINNME inversions: Combining So/PHI and HMI”, 2026, ApJL, in prep.
2. Rivera, Y., ..., Molnar, M., ..., 2025, “2024 Coordinated Eclipse observations: Making a heliospheric energy budget”, *Nat. Astronomy*, under review
3. Molnar, M., Plowman, J., ..., 2025, “Abundance analysis of the M7.6 Flare Observed on Sept 30 2024: A tale of multithermal plasma abundance evolution”, *Astrophys. J.*, under review
4. Pistelli, C., Molnar, M., de Toma, G., 2026, “Observations of eruptive prominences in the middle corona with UCoMP in He I 1083”, *Solar Physics*, under review
5. R. Patel, Molnar, M., Seaton, D., ..., 2026, “Unprecedented Flare Ribbon fine structuring observed with Solar Orbiter”, *Astrophys. J.*, under review
6. S. Shaver, ..., M. Molnar, ..., 2026, *Frontiers in Astronomy and Space Sciences*, “Under One Arkansas Sky: Bridging Science and Community in Rural America”, 13, 1755162
7. Molnar, M., Gilly, C., Morton, R., ..., 2026, “Wave observations with the Cryo-NIRSP instruments at high resolution”, *Astrophys. J.*, 999, 34

8. Plowman, J., Hassler, D., **Molnar, M.**, . . . , 2026, “A new method of deriving Doppler velocities for Solar Orbiter SPICE”, *A&A*, 706, 171
9. Varesano, T., Hassler, D., Zambrana Prado, N., Laming, J. M., Plowman, J., **Molnar, M.**, Barczynski, K., 2026, “FIP Bias in an Emerging Active Region as observed in SPICE Synoptic Observation”, *A&A*, 706, 155
10. **Molnar, M.**, Casini, R., Bryans, P., Berkey, B., Tyson, K., 2025, “Neutral Helium detection during the 2024 Solar Eclipse”, 2025, *Solar Physics*, 300, 88
11. Jarolim, R., **Molnar, M.**, Tremblay, B., Centeno, R., Rempel, M., 2025, “PINN ME: A Physics-Informed Neural Network Framework for Accurate Milne-Eddington Inversions of Solar Magnetic Fields”, *Astrophys. J. Lett.*, 985, L7
12. Morton, R., **Molnar, M.**, Cranmer, S., Schad, T., 2025, “First DKIST Cryo-NIRSP wave observations”, *Astrophys. J.*, 982, 104
13. **Molnar, M.**, Casini, R., 2024, “Magnetic Diagnostics of Prominence Eruptions through the Hanle Effect of the He I 1083 nm Line”, *Astrophys. J.*, 977, 1
14. da Silva Santos, J.M., **Molnar, M.**, Milić, I., de la Cruz Rodriguez, J., Reardon, K., 2024, “Constraining the balance of radiative losses and acoustic flux in the solar chromosphere with multi-line spectral inversions”, *Astrophys. J.*, 976, 21
15. Jarolim, R., Tremblay, B., Rempel, **Molnar, M.**, Veronig, A., Thalmann, J., Podladchikova, T., 2024, “Multi-height magnetic field extrapolations in the solar corona”, *Astrophys. J. Lett.*, 963, 1, id.L21, 13 pp.
16. Cranmer, S., **Molnar, M.**, 2023, “MHD Wave Mode Conversion in the Solar Corona”, *Astrophys. J.*, 955, 68
17. **Molnar, M.**, Cranmer, S., Reardon, K., Kowalski, A., Milić, I., 2023, “Constraining the Systematics of (Acoustic) Wave Heating Estimates in the Solar Chromosphere”, *Astrophys. J.*, 945, 154
18. Tarr, L., Jaeggli, S., Kobelski, A., **Molnar, M.**, Cauzzi, G., Reardon, K., ”Spatio-temporal comparisons of the hydrogen-alpha line width and ALMA 3 mm brightness temperature in the weak solar network”, 2023, *Front. Astron. Space Sci.* 9:978405, doi: 10.3389/fspas.2022.978405
19. Hoffmann, R., Reardon, K., Milić, I., **Molnar, M.**, Chai, Yi, Uitenbroek, H., ”Spectral inversions with IBIS and ALMA”, 2022, *Astrophys. J.*, 933, 244
20. **Molnar, M.**, Cranmer, S., Reardon, K., Kowalski, A., Chai, Y., Gary, D., 2021, “High-frequency wave power observed in the chromosphere with IBIS and ALMA”, *Astrophys. J.*, 920, 125
21. **Molnar, M.**, Reardon, K., Osborne, C., Milić, I., 2020, “Spectral deconvolution with deep learning: removing the effects of spectral PSF broadening”, *Front. Astr. Space Sci.*, 7 (29), doi:10.3389/fspas.2020.00029
22. Sterling, A., Moore, R., Navdeep, P., Reardon, K., **Molnar, M.**, et al., 2020, ”Hi-C 2.1 Observations of Small-scale Miniature-filament-eruption-like Cool Ejections in an Active Region Plage“, *Astrophys. J.*, 889 (2), 187
23. **Molnar, M.**, Reardon, K., Chai, Y., Cauzzi, G., Uitenbroek, H., Gary, D., Cranmer, S., 2019, “Solar Chromospheric Temperature Diagnostics: A Joint ALMA-H α Analysis”, *Astrophys. J.*, 881, 99
24. Kalomeni, B., Nelson, L., Rappaport, S., **Molnar, M.**, et al., 2016, “Evolution of Cataclysmic Variables and Related Binaries Containing a White Dwarf”, *Astrophys. J.*, 833, 17

WHITE PAPERS AND OTHER NON-REFEREED PUBLICATIONS

1. Y. J. Rivera, . . . , **M. Molnar**, . . . , 2024, Solar Orbiter Nuggets, “Coordinated Coronal and Heliospheric Observations During the 2024 Total Solar Eclipse”
2. **M. Molnar**, IRIS Scientific Nugget, May 2023, “Constraining the acoustic flux in the solar atmosphere with UV observations and cutting edge MHD simulations”, (link)
3. K. Reardon, . . . , **M. Molnar**, et al., “Spectroscopic and polarimetric inversions: Our key to unlocking the secrets of the solar atmosphere”, 2023. *Decadal Survey for Solar and Space Physics*

(*Heliophysics*) 2024-2033 white paper e-id. 335; Bulletin of the American Astronomical Society, Vol. 55, No. 3, e-id. 335 (2023)

4. D. Gary, ..., **M. Molnar**, et al., “Frequency Agile Solar Radiotelescope ”, 2023. *Decadal Survey for Solar and Space Physics (Heliophysics) 2024-2033* white paper e-id. 123; Bulletin of the American Astronomical Society, Vol. 55, No. 3, e-id. 123 (2023)

PRESENTATIONS AND SEMINARS

1. Case Western Universit Physics Colloquium, IL, November 2025 (**invited talk**);
2. National Solar Observatory Colloquium, Boulder, September 2025, “Coronal Plasma Abundance Evolution during the M7.6 Flare observed on September 30 2024”, (**invited talk**);
3. Solar Polarization Workshop 11, Prague, September 2025, “Spectropolarimetry of the He I 1083: using the unsaturated Hanle effect for coronal magnetometry”;
4. NLTE Inversions Workshop, University of Glasgow, August 2025, “Spectropolarimetry of the He I 1083: using the unsaturated Hanle effect for coronal magnetometry”;
5. 2nd 2024 Solar Eclipse Workshop, CfA, Boston, August 2025, “SPICE Observations supporting the 2024 Eclipse”;
6. “Analysis of the Solar Plasma Abundance Evolution During the M7.7 Flare Event of September 30, 2024”, AAS/SPD June 2025 meeting, 127.03;
7. T. Schad, ..., **M. Molnar**, ..., AAS/SPD June 2025 meeting, 113.04, “Solar coronal magnetic fields and Alfvénic waves as measured by DKIST/CryoNIRSP”;
8. Institute of Astronomy Colloquium, Bulgarian Academy of Sciences, Sofia, April 2025, Bulgaria, “Spectropolarimetry across the solar atmosphere – Polarization as a solar magnetometer”;
9. SPICE Instrument Consortium meeting, Royal Observatory of Belgium, Brussels, “Studying the elemental abundance evolution during the Sept 30 2024 flare with SPICE and EUV-HRI”;
10. Machine Learning and Computer vision in Heliophysics, Sofia, Bulgaria, April 2025, “Milne-Eddington Spectropolarimetric inversions with PINNs”;
11. CfA Eclipse Workshop, Boston MA, February 2025, “Coronal He I 10830 observations during the total solar eclipse: evidence for terrestrial origin”;
12. AGU Fall Meeting, Washington DC, December 2024, SH54B-07: “Detection of Helium with the CHEESE eclipse experiment”;
13. AGU Fall Meeting, Washington DC, December 2024, NG24A-08: “Spectropolarimetric inversions with Physics Informed Neural Networks”;
14. WHPI Mini-Eclipse workshop, NSF UCAR, November 2024, “Coronal He I 10830 observations during the total solar eclipse”;
15. SwRI Solar Show-and-Tell, November 2024, “Spectropolarimetry through the solar atmosphere”;
16. NSF NCAR/HAO Colloquium, October 2024, “Coronal magnetometry through the He I 1083 Hanle effect”;
17. IMPRS Solar Physics seminar, Göttingen, Germany, September 2024, “Physics Informed Neural Networks for Milne-Eddington inversions”;
18. Deep Learning Workshop at the Rosseland Centre for Solar Physics (RoCS), September 2024, “Physics Informed Neural Networks for Milne-Eddington inversions”;
19. Boulder Solar Focus meeting, May 2024, “Wave behavior observed in the corona with Cryo-NIRSP” (**invited talk**);
20. Boulder Solar Focus meeting, November 2023, “Heating of the solar atmosphere from the photosphere to the corona” (**invited talk**);
21. UCoMP workshop, HAO, Boulder, August 2023, “Coronal He I 1083.0 nm magnetometry”;
22. Leibniz Institute for Solar Physics (KIS), Freiburg, Germany, July 2023, “Wave diagnostics in observations and 3D rMHD models”;
23. IMPRS, Göttingen, Germany, July 2023, “Comparing wave diagnostics in observations and 3D rMHD models”;

24. Harvard/CfA SSP seminar, June 2023, “Solar inversions for stellar applications”;
25. RoCMI workshop, February 2023, Svalbard, Norway, “Constraining the acoustic wave flux in the solar chromosphere with observations and simulations”;
26. NSO Colloquium, November 2022, “Wave heating of the solar chromosphere – systematics and observations”;
27. NAO Colloquium, Institute of Astronomy, Bulgaria, August 2022, “Science with DKIST” **(invited talk)**;
28. TESS Meeting, “Wave heating of the solar chromosphere – systematics on the modeling approaches”, August 2022;
29. Meeting of ALMA Young Astronomers, ESA (virtual), March 2022, “Chromospheric dynamics observed with ALMA”;
30. HAO colloquium, November 2021, “Constraining the wave heating in the solar chromosphere”;
31. Hinode-14/IRIS-11 meeting, October 2021, “High frequency wave heating constrained with IRIS”;
32. NSO Colloquium, September 2021, “Wave heating of the solar chromosphere”;
33. IAGA, August 2021, “High frequency waves in the solar atmosphere observed with IRIS, IBIS and ALMA” **(invited talk)**;
34. AAS/SPD meeting, June 2021, IPoster talk, “Wave heating in the Quiet Solar Chromosphere”;
35. Cool Stars 20, Boston, July 2018, “DKIST Synergies with ALMA” **(invited talk)**;
36. IRIS-9 meeting, MPS, Göttingen, Germany, June 2018, “Probing the turbulent quiet chromosphere with ALMA, IRIS and IBIS”.

POSTER PRESENTATIONS

1. Patel, R., **M. Molnar**, ..., AAS/SPD June 2025 meeting, 307.11, “Unprecedented Fine Structures in Coronal Flare Ribbons Revealed by Solar Orbiters High-Resolution Imager”;
2. J. Plowman, D. Hassler, **M. Molnar**, T. Varesano, AAS/SPD June 2025 meeting, Anchorage, AK, “Updated Correction of Solar Orbiter SPICE Doppler PSF artifacts”;
3. **M. Molnar**, R. Jarolim, R. Centeno, M. Rempel, SDO 2025 meeting, Boulder, CO, “SPINN-ME: Spherical Physics Informed NN Milne-Eddington Inversions”;
4. da Silva Santos, J., **M. Molnar**, AGU Fall meeting 2024, ..., SH21G-2902: “Evaluating acoustic wave flux contributions to spatially resolved chromospheric radiative losses”;
5. Madsen, C., ..., **M. Molnar**, ..., AGU Fall meeting 2024, SH51E-2936: “Characterizing the Coronal Plasma Environment during the 2024 North American Eclipse”;
6. R. Jarolim, **M. Molnar**, et al., Hinode/IRIS meeting, Boulder Solar Day, CO, September, “Milne-Eddington magnetic field inversions with Physics-Informed Neural Networks”
7. **M. Molnar**, R. Casini, SHINE, August 2024, Juneau, AK, “Coronal magnetometry through the He I 1083 nm Hanle-effect polarization”
8. R. Jarolim, **M. Molnar**, et al., Hinode/IRIS meeting, Bozeman, MO, July 2024, “Milne-Eddington magnetic field inversions with Physics-Informed Neural Networks”
9. **M. Molnar**, R. Casini, SpiroFest, Boulder, CO, “Measuring the coronal magnetic field with the Hanle signatures in He I 1083 nm: observational degeneracies and future prospects”;
10. **M. Molnar**, R. Casini, AAS/SPD, August 2023, Minneapolis, “Measuring the coronal magnetic field with the Hanle signatures of He I 1083 nm”;
11. **M. Molnar**, J. da Silva Santos, M. Rempel, K. Reardon, B. Fleck, AAS/SPD, August 2023, Minneapolis, “Explaining the vanishing phases of high frequency wave disturbances in the chromosphere with 3D rMHD models”;
12. L. Tarr, A. Kobelski, S. Jaeggli, **M. Molnar**, G. Cauzzi, K. Reardon, Hinode-15/IRIS-12 meeting, September 2022, Prague, Czech Republic, “Spatio-Temporal comparisons of the Hydrogen Balmer-alpha line width and ALMA 3mm brightness temperature in weak network”
13. **M. Molnar**, K. Reardon, S. Cranmer, Hinode-15/IRIS-12 meeting, September 2022, Prague, Czech Republic, “Center-to-limb variation of chromospheric fluctuations observed with IRIS”

14. **M. Molnar**, K. Reardon, S. Cranmer, SHINE meeting, June 2022, Honolulu, Hawaii, “High frequency wave power in the chromosphere – comparison of observations with 3D models”
15. **M. Molnar**, K. Reardon, Cranmer, S., AAS/SPD meeting, July 2021, online, “Spectroscopic Study Of Wave Propagation In The Quiet Solar Chromosphere with IRIS”, (link)
16. Cranmer, S., Aarnio, A., **Molnar, M.**, AAS/SPD meeting, July 2021, “Chromospheric and Coronal Heating in Cool Stars: Constraints on Physical Processes from X-ray/Ly- α Data” (link)
17. **M. Molnar**, K. Reardon, S. Cranmer, A. Kowalski, AGU, December 2020, online, “High-frequency wave power observed in the chromosphere with IRIS, IBIS and ALMA” (link)
18. **M. Molnar**, K. Reardon, S. Cranmer, SPD meeting, August 2020, online, “High-frequency wave power observed in the chromosphere with IBIS and ALMA” (link)
19. **M. Molnar**, K. Reardon, C. Osborne, I. Milić, ML in Helio, 2019, Amsterdam, Netherlands, “Recovering spectral information with CNNs” (link)
20. **M. Molnar**, K. Reardon, S. Cranmer, SHINE 2019, Boulder, “High frequency chromospheric observations with IBIS and ALMA”
21. **M. Molnar**, K. Reardon, Yi Chai, SHINE 2017, Quebec, Canada, “Exploratory study of the chromosphere with ALMA and IBIS”

SELECTED ACCEPTED OBSERVING PROPOSALS

1. GST/BBSO, **Co-I**, 2023, “Towards an understanding of sources of heating in plage regions”
2. DKIST, **Co-I**, 4 hours, 2023, “Towards an understanding of sources of heating in plage regions”
3. DKIST, **PI**, 4 hours, 2023, “Observing coronal Alfvén-like waves with Cryo-NIRSP”
4. ALMA, **Co-I**, 2021, 4 hours, “Joint ALMA and CO observations”
5. GST/BBSO, **PI**, 2021, 4 hours, “High-frequency waves throughout the lower solar atmosphere”
6. DKIST, **PI**, 2021, 2 hours, “High-frequency waves throughout the lower solar atmosphere”
7. ALMA, **Co-I**, 2019, 4 hours: “Probing the thermal properties of the chromosphere at high resolution”
8. DST/IBIS, **PI**, 2018, 6 observing days: “Polarimetric observations in the K I 7699 Å line”; “He I 10830 Å Coronal hole observations: TR outflows as the roots of the fast solar wind?”
9. ALMA, **Co-I**, 2018, 4 hours: “Investigating the thermal diagnostics of the solar chromospheric network”

NUMERICAL AND LANGUAGE SKILLS

Languages

Bulgarian (native), English (fluent), French (intermediate), Russian (beginner)

Numerical Skills

Python, C/C++, IDL, OpenMP, MPI, scikit-learn, PyTorch, Mathematica

GRANTS, ACADEMIC AWARDS, AND SCHOLARSHIPS

- **NSF SHINE Grant 2401069**
PI, \$394k, October 2024 – October 2026
- **Advanced Study Program Postdoctoral Fellowship**
NCAR Postdoctoral fellowship, ~\$300k, October 2022 – September 2024
- **Chance Irick Cooke Fellowship**
\$3,000, University of Colorado, Boulder, 2022
- **Future Investigators in NASA Earth and Space Science and Technology (FINESST)**
NASA Fellowship, \$90k, September 2020 – August 2022
- **DKIST Ambassador Graduate Fellowship**
National Solar Observatory Fellowship, \$110k, August 2019 – August 2022
- **Hale Graduate Fellowship**
University of Colorado, Boulder, \$160k, August 2016 – August 2019

SERVICE AND OUTREACH

- Advised REU Student Chloe Pistelli, Taylor University, May-June 2025;
- DKIST Time Allocation Committee Panelist, July/August 2025;
- NASA FINESST Proposal Review Panelist, June 2025;
- Co-organized session at the DASH conference, Laurel, MD, October 2023 – “Coronal software tools for data reduction, forward synthesis and inverse problems”;
- Co-organized Interdisciplinary NCAR postdoc retreat, Breckenridge, CO, August 2023;
- Member of the NCAR Advanced Study Program Postdoc Selection committee;
- Reviewer for the DKIST Scientific Proposal Review Committee – 2024, 2025;
- Reviewer for the ApJ, ApJ Letters, MNRAS, Astronomy & Astrophysics, Solar Physics;
- Presenter at Skype-a-Scientist initiative, Spring 2018 – present;
- Lecturer and Activity leader at “Beli Brezi” Astronomical Summer School, Bulgaria – 2015, 2017, 2019, 2022, 2023;
- Problem author for the Bulgarian National Astronomical Olympiad – 2022;
- Organizer of the NSO/APS Graduate Solar Journal club Spring 2020 – Summer 2022;
- Advised Chase Parker (high school) on Senior Honors project, Fall 2018 – Spring 2019.